DIAGONAL REFLECTION POSITIVITY AND ROTATION INVARIANCE WITHOUT CLOSEDNESS

Abstract. A Gaussian-dominated law of distributional two-forms on $\mathbb{R}^3$, invariant under translations and the cubic group and reflection positive at the coordinate planes, is Euclidean invariant if it is reflection positive at one diagonal plane. All its L^2 moments are covariant, including at overlapping supports. We also prove a two-point result under weaker symmetry assumptions, compare the angular criteria for scalar and field-strength kernels, and give examples separating diagonal positivity from the spectral cone and closedness.

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1. Introduction

Let μ be a probability law on the real tempered two-forms $\mathcal{S}'(\mathbb{R}^3;\Lambda^2)$, with the sigma-algebra generated by evaluation on real Schwartz tests. Let F be the canonical random distribution and put

$$*f = (f_{23}, -f_{13}, f_{12}), \qquad \mathbb{B}(b) = F(*^{-1}b).$$

For an isometry $\mathfrak{g}(x) = gx + a$ we use

$$(\mathfrak{g}\cdot b)(x) = \det(g)\, g\, b(\mathfrak{g}^{-1}x), \qquad (\mathfrak{g}F)(f) = F(\mathfrak{g}^{-1}\cdot f).$$

Here the action on two-form tests is $(\mathfrak{g}\cdot f)(x) = (\wedge^2 g) f(\mathfrak{g}^{-1}x)$. Reflection positivity has the characteristic-functional meaning in (1) below, on tests compactly supported in the open positive half space.

Theorem A. Suppose that μ has the following properties:

(i) Gaussian domination with constant one:

$$\mu(e^{\mathbb{B}(b)}) \le e^{\|b\|_2^2/2} \quad \text{for every real Schwartz vector test } b; \tag{G}$$

(ii) invariance under all translations and under the signed permutation group $O_{\mathrm{cub}} = O(3)\cap \mathrm{GL}(3,\mathbb{Z})$;

(iii) reflection positivity at $\{x_j = 0\}$ for $j = 1,2,3$, and at one plane $\{x_i = \sigma x_l\}$, where $i < l$ and $\sigma \in \{1,-1\}$, from either chosen side.

Then $\mathfrak{g}_*\mu = \mu$ for every Euclidean isometry $\mathfrak{g}$. For every $N \ge 1$ the moment

$$S_N(b_1,\dots,b_N) = \mu\big(\mathbb{B}(b_1)\cdots\mathbb{B}(b_N)\big)$$

extends to a bounded complex multilinear form on $L^2(\mathbb{R}^3;\mathbb{C}^3)^N$ and satisfies

$$S_N(\mathfrak{g}\cdot b_1,\dots,\mathfrak{g}\cdot b_N)=S_N(b_1,\dots,b_N)$$

for all such tests, without a support restriction. ■□

Hypothesis (H) means Gaussian domination, translation and cubic invariance, and reflection positivity at the three coordinate planes. The additional hypothesis of the theorem is positivity at one diagonal plane. No condition $dF=0$ is imposed.

There are two obstacles to deriving the theorem from the spectral cone. At two points, the angular growth estimate permits one extra frequency for each of the equal circular labels. The law turned through $\pi/4$ has the same analytic bounds, while this extra coefficient has the opposite sign in its reflection form. Positivity in both frames removes the coefficient. The higher-order harmonic argument then proves covariance at distinct planar points.

The second obstacle is contact. The moments exist on all L^2 tests, so covariance away from coincidences does not yet prove the theorem. We localize two entries in normalized discs of radius d. Gaussian domination bounds their averaged products uniformly as $d\downarrow 0$. Weak limits of the associated pair fields define bounded fields over the plane. The cone makes their two-point kernels holomorphic, and the quarter turn joins the four angular charts into a bounded entire function. Liouville's theorem forces their reconstructed vectors into the zero-energy subspace. In an ergodic component that subspace is the vacuum line. A further covariance estimate is needed to show that the fields themselves are constant in $L^2(\mu)$; equality of their reconstructed classes would not suffice. This gives factorization of a contact pair. Induction then removes the disjointness conditions one pair at a time.

The analytic tools used before this contact argument are the cone, plate-kernel and harmonic arguments in [5, Sections 7–8]. We recall the exact forms needed below. Their hypotheses are conditions on the law; the proof here does not require a lattice action or a choice of renormalization parameters. The proof of theorem A is completed in section 5, where the covariance proved on the ergodic components is integrated and the other coordinate planes are treated by cubic symmetry.

1.1. **Reflection forms, moments and the cone.** For a unit vector n and $c\in\mathbb{R}$, let $r_{n,c}x=x-2(n\cdot x-c)n$. Reflection positivity means

$$\sum_{a,b=1}^{k}\overline{z_a}z_b\,\mu\Big(e^{i\{F(f_b)-F(r_{n,c}\cdot f_a)\}}\Big)\ge 0 \tag{1}$$

for every $k\ge 1$, $z\in\mathbb{C}^k$, and real smooth two-form tests f_a with compact support in $\{n\cdot x>c\}$. We denote its left side by $\mathcal{Q}_\mu(n,c;z;f)$. For a law invariant under $r_{n,c}$, the two choices of normal give equivalent conditions. Reflection invariance is stated separately whenever it is needed.

Write $\theta_j=r_{e_j,0}$. Conditions (T), (R_j) and (RP_j) will mean translation invariance, invariance under θ_j, and positivity at $x_j=0$, respectively. For rotations set

$$R_t^{ji}e_j=\cos t\,e_j+\sin t\,e_i,\qquad R_t^{ji}e_i=-\sin t\,e_j+\cos t\,e_i.$$

The third basis vector is fixed. Condition (Q_{ji}) means invariance under $R^{ji}_{\pi/2}$. For a law with (T), (R_j) and (RP_j), the reflection-positive reconstruction has a class map $[\,\cdot\,]_j:L^2(\Sigma_j^+,\mu)\longrightarrow\mathcal{H}_j$ with scalar product

$$\langle[\Phi]_j,[\Psi]_j\rangle=\mu((\Theta_j\Phi)\Psi). \tag{2}$$

Here Σ_j^+ is the sigma-algebra generated by evaluations on real smooth tests compactly supported in $\{x_j>0\}$, and $\Theta_j\Phi=\overline{\hat\theta_j\Phi}$, with $(\hat{\mathfrak{g}}\Phi)(F)=\Phi(\mathfrak{g}^{-1}F)$. The map is a contraction and has dense range; $\Omega_j=[1]_j$. The nonnegative time generator H_j and the commuting transverse momenta have joint spectral measure E_j. We write (C_{ji}) for the support condition

$$E_j\{(\omega,p_\perp):\omega<|p_i|\}=0,\quad\text{equivalently } H_j\ge|P_i|. \tag{3}$$

The cone theorem [5, Proposition 7.2] says that $(T), (R_j), (RP_j)$, together with reflection invariance and positivity at the plane with normal $(e_j + e_i)/\sqrt{2}$, imply (C_{ji}). It assumes no closedness. In the $(1,2)$-plane we abbreviate

$$(G),\ (T),\ (R_1),\ (RP_1),\ (Q_{12}),\ (C_{12}). \tag{S}$$

For $1 \le p < \infty$ and integers $N \ge 1$, Gaussian domination implies the bounds

$$\|\mathbb{B}(b)\|_{L^p(\mu)} \le c_p \|b\|_2, \qquad |S_N(b_1, \dots, b_N)| \le c_N^N \prod_a \|b_a\|_2, \qquad S_1 = 0, \tag{4}$$

and $c_2 = 1$. These statements include the unique continuous extension to L^2 tests. In particular,

$$\|\mathbb{B}(b)\mathbb{B}(b')\|_{L^2(\mu)} \le c_4^2 \|b\|_2 \|b'\|_2. \tag{5}$$

These bounds and the entire finite-dimensional generating functions follow from [5, Proposition 5.1]. The Taylor coefficients determine those generating functions, hence the characteristic functions of all finite families of evaluations. Thus invariance of every moment on real Schwartz tests is equivalent to invariance of the law. Under a symmetry of the law, the action on $\mathbb{B}$ extends to L^2 tests, and moment invariance extends by (4).

Let $\mathcal{C}_9$ denote the translation-invariant laws that are invariant and reflection positive at every plane in the nine families $x_j = c$ and $x_i \pm x_l = c$, $i < l$, $c \in \mathbb{R}$.

Lemma 1.1 (Transport of diagonal positivity)**.** Under (H), positivity at one diagonal plane through the origin is equivalent to membership in $\mathcal{C}_9$. ■

Proof. Each coordinate or diagonal reflection through the origin is a signed permutation. Hence (H) gives invariance under all of them. The signed permutation group acts transitively on the twelve unit vectors $(\pm e_i \pm e_l)/\sqrt{2}$, $i < l$. Conjugation gives

$$g r_{n,0} g^{-1} = r_{gn,0}.$$

Changing variables in (1) therefore transports positivity from the chosen diagonal normal to every such normal. Coordinate positivity is already in (H). Finally,

$$r_{n,c} = \tau_{cn} r_{n,0} \tau_{-cn}$$

and translation invariance transport both positivity and reflection invariance to every parallel plane. This is precisely $\mathcal{C}_9$. The converse follows from its definition. □

2. Two-point covariance with one diagonal mirror

The two-point assertion needs less symmetry than theorem A.

Write $x = (y, x_3)$, $y = (y^0, y^1) = (x_1, x_2)$, $R_t = R_t^{12}$ and $n_\upsilon = (\cos\upsilon, \sin\upsilon)$, also read as a vector in $\mathbb{R}^3$. Set

$$e^{\pm} = e_1 \pm i e_2, \quad e^3 = e_3, \qquad w(\pm) = \pm 1, \quad w(3) = 0, \qquad s(\pm) = 1, \quad s(3) = -1.$$

A label $\lambda = (\sigma, \chi)$, with real $\chi \in L^2(\mathbb{R})$, has sign $s(\lambda) = s(\sigma)$ and, for $\alpha \in L^2(\mathbb{R}^2; \mathbb{C})$, specifies the plate $(\alpha \otimes \lambda)(x) = \alpha(y)\chi(x_3)e^{\sigma}$.

Theorem B. Suppose μ satisfies $(G), (T), (R_1), (RP_1), (Q_{12})$ and is reflection positive at $x_1 = x_2$ or at $x_1 = -x_2$, from one chosen side. Then

$$S_2(R_t^{12} \cdot b, R_t^{12} \cdot b') = S_2(b, b') \quad (t \in \mathbb{R},\ b, b' \in L^2(\mathbb{R}^3; \mathbb{C}^3)).$$

For labels $\lambda = (\sigma, \chi)$ and $\lambda' = (\sigma', \chi')$ with real $\chi, \chi' \in C_c^\infty(\mathbb{R})$, the two-point plate kernel has the form

$$G_{\lambda\lambda'}(r n_\vartheta) = a_{\lambda\lambda'}(r) e^{iw\vartheta}, \qquad |a_{\lambda\lambda'}(r)| \le \frac{72}{\pi} \|\chi\|_2 \|\chi'\|_2 r^{-2},$$

for every $r > 0$ and $\vartheta \in \mathbb{R}$, where $w = w(\sigma) + w(\sigma')$. The kernel represents S_2 on compact, disjoint planar supports as in (14). ■□

2.1. **The angular criterion.** For scalar kernels, rotation invariance means independence of the angle. For two-form kernels, the vector labels also rotate, and covariance requires a prescribed phase. The following criterion separates this finite-dimensional calculation from the construction of the kernels. Its second part will apply after the kernels have been joined.

Proposition 2.1 (Elimination of the remaining angular modes)**.** The following assertions hold.

(i) Let h be entire and suppose, for some finite $A \geq 0$, that

$$h(\zeta + \pi/2) = h(\zeta), \qquad |h(t + i\beta)| \leq Ae^{4|\beta|} \quad (t, \beta \in \mathbb{R}).$$

If $h(i\beta) \geq 0$ and $h(\pi/4 + i\beta) \geq 0$ for every $\beta \in \mathbb{R}$, then h is a nonnegative constant.

(ii) Let $w \in \{-2, -1, 0, 1, 2\}$ and let h be entire. Suppose, for some finite $A \geq 0$, that

$$h(\zeta + \pi/2) = e^{iw\pi/2} h(\zeta), \qquad |h(t + i\beta)| \leq Ae^{2|\beta|} \quad (t, \beta \in \mathbb{R}).$$

For $w = \pm 2$, assume in addition that

$$h(i\beta) \geq 0, \qquad e^{-iw\pi/4} h(\pi/4 + i\beta) \geq 0 \quad (\beta \in \mathbb{R}).$$

Then $h(\zeta) = ae^{iw\zeta}$ for some $a \in \mathbb{C}$. For $w = \pm 2$, one has $a \geq 0$.

Each inequality includes the assertion that its left side is real. ■

The following Fourier coefficient argument is the one used in [5, Section 8.5], with an arbitrary growth exponent q.

Lemma 2.2. Let $w \in \mathbb{Z}$ and $q \geq 0$. Suppose that h is entire and that, for some finite $A \geq 0$,

$$h(\zeta + \pi/2) = e^{iw\pi/2} h(\zeta), \qquad |h(t + i\beta)| \leq Ae^{q|\beta|} \quad (t, \beta \in \mathbb{R}).$$

Then

$$h(\zeta) = \sum_{\substack{k \in \mathbb{Z},\ |k| \leq q \\ k - w \in 4\mathbb{Z}}} a_k e^{ik\zeta}, \qquad |a_k| \leq A.$$

■

Proof. The function h is 2π-periodic. For $k \in \mathbb{Z}$, its Fourier coefficient can be written

$$a_k = \frac{1}{2\pi} \int_0^{2\pi} h(t + i\beta) e^{-ik(t+i\beta)}\, dt.$$

The integral is independent of β: integrate the entire periodic integrand around a rectangle and cancel the vertical sides. Hence $|a_k| \leq Ae^{q|\beta| + k\beta}$. Letting $\beta \to -\infty$ for $k > q$ and $\beta \to +\infty$ for $k < -q$ removes those coefficients. Translation by $\pi/2$ gives $a_k = 0$ unless $k - w \in 4\mathbb{Z}$. Fourier uniqueness on the real axis and the identity theorem give the finite expansion; $\beta = 0$ gives its coefficient bound. □

Proof of proposition 2.1. In the scalar case, lemma 2.2 leaves $h(\zeta) = c_0 + c_1 e^{4i\zeta} + c_{-1} e^{-4i\zeta}$. Positivity gives

$$c_0 + c_1 e^{-4\beta} + c_{-1} e^{4\beta} \geq 0, \qquad c_0 - c_1 e^{-4\beta} - c_{-1} e^{4\beta} \geq 0.$$

Divide by $e^{4\beta}$ and let $\beta \to +\infty$ to obtain $c_{-1} \geq 0$ and $-c_{-1} \geq 0$. Thus $c_{-1} = 0$. Division by $e^{-4\beta}$ and passage to $\beta \to -\infty$ give $c_1 = 0$. The remaining inequality gives $c_0 \geq 0$.

In the tensor case, the conditions $|k| \leq 2$ and $k - w \in 4\mathbb{Z}$ leave only $k = w$, except when $w = \pm 2$. In those cases write $h(\zeta) = a^w e^{iw\zeta} + a^{-w} e^{-iw\zeta}$. Since $e^{-2iw\pi/4} = -1$, positivity reads

$$a^w e^{-w\beta} + a^{-w} e^{w\beta} \geq 0, \qquad a^w e^{-w\beta} - a^{-w} e^{w\beta} \geq 0.$$

Multiply by $e^{-w\beta}$ and let $w\beta \to +\infty$. The two limits give $a^{-w} \geq 0$ and $-a^{-w} \geq 0$. Thus $a^{-w} = 0$, and the first inequality then gives $a^w \geq 0$. □

For a scalar plate kernel, such as a kernel associated with $\operatorname{tr} F \cdot F$, the first part of proposition 2.1 applies once the following analytic inputs have been established: the short-distance bound $t^8\Gamma(t) \to 0$, where Γ is the Laplace transform of the energy marginal of a positive joint energy–momentum measure; a contour argument leaving only the frequencies $-4, 0, 4$; domination $\widetilde{\rho}_{\mathrm{d}} \le C\widetilde{\rho}$ of the diagonal joint measure by a measure supported on the cone; and identification of the continued transforms with $h(i\beta)$ and $h(\pi/4 + i\beta)$. Given these inputs, the first part of proposition 2.1 removes the two nonconstant coefficients. The two-form argument below instead obtains its growth exponent two from Gaussian domination and retains the phase dictated by the vector labels.

2.2. **Plates and the eight oriented mirrors.** Finite sums of plates are dense in $L^2(\mathbb{R}^3;\mathbb{C}^3)$, also with a prescribed measurable support in the plane. Their norms and actions are

$$\|\alpha \otimes \lambda\|_2 \le \sqrt{2}\|\alpha\|_2\|\chi\|_2, \tag{6}$$

$$R_t \cdot (\alpha \otimes \lambda) = e^{-iw(\sigma)t}(\alpha \circ R_{-t}) \otimes \lambda, \tag{7}$$

$$\Theta_1 \mathbb{B}_\lambda(\alpha) = s(\sigma)\mathbb{B}_\lambda(\bar{\alpha} \circ \theta_1). \tag{8}$$

Here $\mathbb{B}_\lambda(\alpha) = \mathbb{B}(\alpha \otimes \lambda)$. The plate identities and their use in the point and ordered-kernel constructions are given in [5, Sections 8.2–8.3]. The density assertion follows by approximating each component by finite sums of products of a planar L^2 function and a transverse L^2 function; the planar factors can be restricted to the prescribed measurable set.

The group generated by θ_1 and $R_{\pi/2}$ contains $r_{n_{\pi/4},0} = R_{\pi/2}\theta_1$ and all its quarter-turn conjugates. Thus the law is invariant under the reflections with normals n_υ, $\upsilon \in (\pi/4)\mathbb{Z}$. Quarter-turn transport of coordinate positivity gives the normals $\upsilon \in (\pi/2)\mathbb{Z}$; transport of the assumed diagonal positivity gives the intervening normals. Both sides of each of the four planes are included.

For each frame $\upsilon \in (\pi/4)\mathbb{Z}$ put

$$\mu_\upsilon = (R_{-\upsilon})_* \mu.$$

Only two laws occur, since $\mu_{\upsilon+\pi/2} = \mu_\upsilon$. In the frame υ, the adjacent normals $n_\upsilon, n_{\upsilon+\pi/4}$ supply coordinate and diagonal reflection positivity. Orthogonal transport preserves (G) and (T) and conjugates the symmetry assumptions. The cone theorem therefore gives (C_{12}) for every μ_υ. Its two-point function obeys

$$S_2^\upsilon(b, b') = S_2(R_\upsilon \cdot b, R_\upsilon \cdot b'). \tag{9}$$

2.3. **Joining the kernels.** Throughout this kernel construction take real $\chi, \chi' \in C_c^\infty(\mathbb{R})$. General L^2 tests enter after the kernel identity has been proved, through plate density and the moment bound.

The one-picture plate-kernel construction in [5, Section 8.4] applies to each μ_υ: its assumptions are $(G), (T), (R_1), (RP_1)$ and the cone on the one-field subspace, which follows from (C_{12}). It gives a smooth kernel $G^\upsilon_{\lambda\lambda'}$ on $\{y^0 > 0\}$ and a holomorphic function $h^\upsilon = h^{\upsilon,r}_{\lambda\lambda'}$ on $|\Re\zeta| < \pi/2$. The statements we use are

$$S_2^\upsilon(\alpha \otimes \lambda, \alpha' \otimes \lambda') = \iint \alpha(y)\alpha'(y')G^\upsilon_{\lambda\lambda'}(y' - y)\, dy\, dy', \tag{10}$$

$$h^\upsilon(\vartheta) = G^\upsilon_{\lambda\lambda'}(r n_\vartheta), \tag{11}$$

$$|h^\upsilon(t + i\beta)| \le \frac{36}{\pi}\|\chi\|_2\|\chi'\|_2(r\cos t)^{-2}e^{2|\beta|}, \tag{12}$$

$$s(\sigma)h^\upsilon_{\lambda\lambda}(i\beta) \ge 0. \tag{13}$$

The first formula requires compact supports ordered by $\sup_{\operatorname{supp}\alpha} y^0 < \inf_{\operatorname{supp}\alpha'} y^0$; the next two use $|\vartheta|, |t| < \pi/2$. The last holds for every real β.

Fix the two labels and put $w = w(\sigma) + w(\sigma')$. The kernel in the original coordinates of frame υ is

$$G^{(\upsilon)}(y) = e^{iw\upsilon}G^\upsilon_{\lambda\lambda'}(R_{-\upsilon}y), \qquad n_\upsilon \cdot y > 0.$$

By (7) and (9), it represents the original S_2 when the supports are ordered along n_υ. If two frames apply, the resulting integrals agree for all tests in sufficiently small compact neighborhoods of any pair in the common domain. Equality of continuous densities gives $G^{(\upsilon)} = G^{(\upsilon')}$ there. These half planes cover $\mathbb{R}^2 \setminus \{0\}$, so the kernels define a single smooth $G_{\lambda\lambda'}$ on that set.

The density formula holds for every pair of compact, disjoint supports:

$$S_2(\alpha \otimes \lambda, \alpha' \otimes \lambda') = \iint \alpha(y)\alpha'(y')G_{\lambda\lambda'}(y' - y)\, dy\, dy'. \tag{14}$$

To see this, partition both supports into sufficiently small squares. Their positive distance allows each pair of squares to be strictly ordered in one of the four coordinate directions. Apply the ordered formula to each pair and sum; only finitely many squares meet the supports. All integrals converge absolutely because the supports are compact, the tests are in L^1 there, and the kernel is bounded on their set of differences. The density is unique. It is consequently bilinear in the real profiles and satisfies

$$G_{\lambda\lambda'}(y) = G_{\lambda'\lambda}(-y). \tag{15}$$

For a fixed $r > 0$, define on $|\Re\zeta - \upsilon| < \pi/2$

$$h^{(\upsilon)}(\zeta) = e^{iw\upsilon}h^\upsilon(\zeta - \upsilon).$$

On real overlaps these functions equal $G_{\lambda\lambda'}(rn_\vartheta)$. The identity theorem joins them to an entire function h. Choosing a frame with $|t - \upsilon| \le \pi/4$ in (12) gives

$$|h(t + i\beta)| \le \frac{72}{\pi}\|\chi\|_2\|\chi'\|_2 r^{-2}e^{2|\beta|}. \tag{16}$$

Since the chart kernels repeat after $\pi/2$, one also has $h(\zeta + \pi/2) = e^{iw\pi/2}h(\zeta)$. Lemma 2.2 now gives

$$h(\zeta) = \sum_{\substack{|k|\le 2 \\ k - w \in 4\mathbb{Z}}} a^k(r)e^{ik\zeta}. \tag{17}$$

Each coefficient is bounded by the right side of (16) with $\beta = 0$.

2.4. **The sign at the diagonal frame.** Only the equal circular labels $\sigma = \sigma' = \pm$ admit a frequency other than w: then $w = \pm 2$ and the other frequency is $-w$. Start with equal profiles, so $\lambda' = \lambda$. Since $s(\sigma) = 1$, (13) in frame υ reads

$$e^{-iw\upsilon}h(\upsilon + i\beta) \ge 0.$$

The frames 0 and $\pi/4$ give the two positivity assumptions in proposition 2.1. Together with (16) and the quarter-turn relation, its second part gives $a^{-w} = 0$. For unequal profiles the coefficient is bilinear and, by (15), symmetric at the even frequency $-w$. Polarization gives its vanishing for those profiles as well.

It follows that $h(\zeta) = a^w(r)e^{iw\zeta}$ for every pair of labels, and thus

$$G_{\lambda\lambda'}(R_t y) = e^{iwt}G_{\lambda\lambda'}(y). \tag{18}$$

This proves the kernel assertion of theorem B. Substitution in (14) cancels the phase from (7). Hence the rotation defect

$$D_t(b, b') = S_2(R_t \cdot b, R_t \cdot b') - S_2(b, b')$$

vanishes for tests over disjoint compact planar sets, first for plates and then by density. It remains bounded on the whole space:

$$|D_t(b, b')| \le 2\|b\|_2\|b'\|_2. \tag{19}$$

2.5. **The two-point contact defect.** We need the following elementary form of locality.

Lemma 2.3. Let E be a bounded bilinear form on $L^2(\mathbb{R}^2;\mathbb{C})$. If it is translation invariant and vanishes on tests supported in disjoint compact sets, then $E(\alpha,\alpha') = c\int \alpha\alpha'$ for a constant $c \in \mathbb{C}$. ■

Proof. For the dyadic squares $Q_{n,k} = 2^{-n}(k+[0,1)^2)$, the form vanishes on distinct squares. Indeed, shrink each square to a compact interior square, apply the hypothesis, and pass to the limit in L^2. Translation invariance makes $E(1_{Q_{n,k}}, 1_{Q_{n,k}})$ independent of k. Splitting a square into four children shows that this value is $4^{-n}c$, where $c = E(1_{[0,1)^2}, 1_{[0,1)^2})$. The required formula follows on finite dyadic step functions. Such functions are dense in L^2, and both sides are bounded bilinear forms, so it holds everywhere. □

Apply the lemma, for fixed labels and t, to $E_t(\alpha,\alpha') = D_t(\alpha\otimes\lambda, \alpha'\otimes\lambda')$. Translation invariance follows from $R_t\tau_v = \tau_{R_tv}R_t$ and (T). Thus

$$E_t(\alpha,\alpha') = c_t(\lambda,\lambda')\int \alpha\alpha'.$$

Under simultaneous rotation of the two plates this expression acquires the phase e^{-iws}. Quarter-turn invariance of S_2 makes it unchanged at $s=\pi/2$, so $c_t = 0$ unless $w = 0$. Since $|w| \le 2$, these are all the possibilities. In either case D_t is invariant under every simultaneous rotation, first on plates and then by density on all tests.

The rotation defect satisfies the cocycle identity

$$D_{t+s}(b,b') = D_t(R_s\cdot b, R_s\cdot b') + D_s(b,b').$$

Its just-proved invariance makes this $D_{t+s} = D_t + D_s$. For every integer $k \ge 1$, (19) therefore gives

$$k|D_t(b,b')| = |D_{kt}(b,b')| \le 2\|b\|_2\|b'\|_2.$$

Letting $k\to\infty$ proves $D_t = 0$ and completes the proof of theorem B.

2.6. **Comparison with closedness.** There is another way to remove the same exceptional coefficient. Suppose that a law satisfies (S) and is closed, so that $\mathbb{B}(\nabla u) = 0$ for scalar Schwartz tests u. Under these assumptions, the two-point kernel construction in [5, Section 8.5] gives the same frequency window as (17). For real $\chi_0,\chi' \in C_c^\infty(\mathbb{R})$ and $\lambda' = (\sigma',\chi')$, put $z = y^0 + iy^1$ and $\partial_z = (\partial_{y^0} - i\partial_{y^1})/2$. Take $\eta,\alpha' \in C_c^\infty(\mathbb{R}^2;\mathbb{C})$ with disjoint supports. Pairing

$$\nabla(\eta(y)\chi_0(x_3)) = (\partial_z\eta)\otimes(+,\chi_0) + (\partial_{\bar z}\eta)\otimes(-,\chi_0) + \eta\otimes(3,\chi_0')$$

with the plate $\alpha'\otimes\lambda'$, and integrating by parts in the density formula, gives

$$\partial_z G_{(+,\chi_0)\lambda'} + \partial_{\bar z} G_{(-,\chi_0)\lambda'} + G_{(3,\chi_0')\lambda'} = 0 \quad \text{on } \mathbb{R}^2\setminus\{0\}. \tag{20}$$

For a radial coefficient f,

$$\partial_z(fe^{ik\vartheta}) = \tfrac12 e^{i(k-1)\vartheta}(f' + kf/r), \qquad \partial_{\bar z}(fe^{ik\vartheta}) = \tfrac12 e^{i(k+1)\vartheta}(f' - kf/r).$$

When $\sigma' = +$, only the exceptional coefficient $f = a^{-2}_{++}$ contributes to frequency -3 in (20); when $\sigma' = -$, only $f = a^2_{--}$ contributes to frequency 3. In both cases it follows that $f' - 2f/r = 0$, hence $f(r) = Cr^2$. The coefficient bound $|f(r)| \le cr^{-2}$ holds for every $r > 0$, and passage to $r\to\infty$ gives $C = 0$. The use of the estimate at all radii is necessary here.

Closedness and diagonal positivity therefore supply different hypotheses for eliminating the exceptional frequency. Rotational covariance itself does not imply closedness; proposition 6.2 gives an invariant law positive at every plane for which $dF \neq 0$. At coincidences, the bounded-form theorem of [4, Theorem A] requires vanishing both on entries over pairwise disjoint compact planar sets and whenever any entry is a compactly supported gradient. The proof in section 4 uses localized pairs to remove the support restriction without this gradient condition.

3. All moments away from coincidences

We next use the higher-order harmonic constructions and estimates of [5, Sections 8.7–8.9]. Their proofs use exactly (S); the closedness argument at two points is separate. For $N \geq 2$ and a tuple of labels $\lambda = (\lambda_1, \dots, \lambda_N)$, they give real analytic functions on

$$\mathrm{Conf}_N(\mathbb{R}^2) = \{(y_1, \dots, y_N) : y_a \neq y_b \ (a \neq b)\}$$

with finite expansion

$$\widetilde{G}_N(R_t Y; \lambda) = \sum_m e^{i(w+4m)t} \widetilde{c}_m(Y; \lambda), \qquad w = \sum_a w(\sigma_a). \tag{21}$$

The distinct-configuration density argument in [5, Section 8.10] identifies these kernels as the moment densities on plates with compact, pairwise disjoint planar supports. Its proof moves one of two time-separated groups transversely. The cone makes the correlation analytic, and the identity theorem extends the ordered density formula to the configuration space of distinct planar points. This identification is needed in addition to the angular expansion. It does not remove terms supported at coincidences.

For $N \geq 2$, the anomaly index $\mathfrak{a}(N)$ is the largest $|m|$ for which $\widetilde{c}_m(\,\cdot\,; \lambda)$ is not identically zero for some tuple of labels. It is zero if every harmonic vanishes; set $\mathfrak{a}(0) = \mathfrak{a}(1) = 0$. The harmonic and polygon results give, for nonnegative integers a, b and N,

$$\mathfrak{a}(0) = 0, \qquad \mathfrak{a}(N) \leq N/2, \tag{22}$$

$$2\mathfrak{a}(a+b) \leq \mathfrak{a}(2a) + \mathfrak{a}(2b), \tag{23}$$

$$a \mid 2\mathfrak{a}(2a) \quad (a \geq 1). \tag{24}$$

Complex conjugation exchanges the circular labels and replaces the index m by $-m$. Whenever $\mathfrak{a}(N) > 0$, both extreme indices occur. In particular, if $\mathfrak{a}(2) = 1$, some two-point harmonic $\widetilde{c}_{-1}$ is nonzero. The top-harmonic lemma gives the positive semidefinite kernel

$$K_1((x;\lambda),(z;\lambda')) = s(\lambda)\widetilde{c}_{-1}(\theta_1 x, z; \lambda, \lambda'), \qquad x, z \in \{y^0 > 0\}. \tag{25}$$

The harmonic extension, convexity, regular-polygon argument and positive top-coefficient kernel are proved in [5, Sections 8.7–8.9]. These arguments do not use $dF = 0$.

Proposition 3.1. Suppose μ satisfies $(G), (T), (R_1), (RP_1), (R_2), (RP_2), (Q_{12})$ and is invariant and reflection positive at $x_1 + x_2 = 0$. Then $\mathfrak{a}(N) = 0$ for every N, and all its moments are covariant under R_t on tests over pairwise disjoint compact planar sets. ∎

Proof. The cone theorem gives (S) for μ. Let $g = R_{\pi/4}$ and $\mu' = (g^{-1})_*\mu$. The map g sends the coordinate normal e_1 to $n_+ = (e_1 + e_2)/\sqrt{2}$ and sends n_+ to e_2. Thus μ' has the coordinate and diagonal symmetries and positivity needed for the cone theorem as well. It has $(G), (T), (Q_{12})$ by transport, so it too satisfies (S).

The moment transformation formula and uniqueness of the plate densities give

$$\widetilde{G}'_N(Y; \lambda) = e^{-iw\pi/4} \widetilde{G}_N(R_{\pi/4} Y; \lambda).$$

Use (21) and compare the finite Fourier expansions after replacing Y by $R_t Y$. This gives

$$\widetilde{c}'_m = (-1)^m \widetilde{c}_m. \tag{26}$$

If $\mathfrak{a}(2) = 1$, a nonzero $\widetilde{c}_{-1}$ remains nonzero for μ', so its index is also one. The top-harmonic kernel for μ' is then $-K_1$. Both K_1 and $-K_1$ are positive semidefinite, which forces their diagonal and hence the entire kernel to vanish. Thus $\widetilde{c}_{-1}$ vanishes on $(-\mathbb{H}) \times \mathbb{H}$, where $\mathbb{H} = \{y^0 > 0\}$. Real analyticity on the connected configuration space forces its vanishing everywhere, a contradiction. The bound $\mathfrak{a}(2) \leq 1$ therefore gives $\mathfrak{a}(2) = 0$.

For $\nu(a) = \mathfrak{a}(2a)$, (22)– (24) give a nonnegative integer sequence with $\nu(0) = 0$, $\nu(a) \le a$, nondecreasing increments and $a \mid 2\nu(a)$. The integer-sequence argument in [5, Section 8.9] therefore gives $\mathfrak{a}(2a) = a\mathfrak{a}(2)$. For arbitrary N, choose $a = \lfloor N/2 \rfloor$ and $b = N - a$ in (23). Thus

$$\mathfrak{a}(N) \le \frac{N}{2}\mathfrak{a}(2). \tag{27}$$

The vanishing of the two-point index therefore gives $\mathfrak{a}(N) = 0$ for every N.

Consequently (21) reduces to

$$\widetilde{G}_N(R_t Y; \lambda) = e^{iwt}\widetilde{G}_N(Y; \lambda).$$

In the density formula this phase cancels the product of the plate phases in (7). This proves covariance for plates over pairwise disjoint compact sets. Plate density with prescribed supports and (4) give it for all tests over those sets. □

Proposition 3.1 proves the separated-support assertion needed for theorem A. We now remove the support restriction.

4. Contact pairs and invariance on all test fields

The angular argument gives covariance on entries over pairwise disjoint compact subsets of the plane. To remove one disjointness condition, we localize the corresponding pair in the same disc. Gaussian domination gives pair fields bounded from $L^1(\mathbb{R}^2)$ to $L^2(\mu)$, uniformly in the disc radius. A weak limit has a bounded reconstructed two-point kernel on the forward tube. Quarter-turn covariance and Liouville's theorem put its reconstructed classes in the zero-energy space. When that space is the vacuum line, reflection factorization and an L^1 strip estimate show that the field itself is constant in $L^2(\mu)$. This yields factorization of the localized pair; induction then removes the disjointness conditions one pair at a time.

Throughout this section write

$$\mathfrak{h} = L^2(\mathbb{R}^3; \mathbb{C}^3), \qquad x = (y, x_3), \quad y = (x_1, x_2), \qquad R_t = R_t^{12},$$

and let $R = R_{\pi/2}$. The assumptions (S) are those of (S). The unadorned norm on $\mathfrak{h}$ is its L^2 norm. The additional condition $\ker H_1 = \mathbb{C}\Omega_1$ will be stated where it is used. Here $\mathcal{H}_1$, $[\,\cdot\,]_1$, H_1 and $\Omega_1 = [1]_1$ are the reconstruction in the first coordinate. For a Borel set $K \subset \mathbb{R}^2$ set

$$\mathfrak{h}_K = \{f \in \mathfrak{h} : f = 0 \text{ a.e. outside } K \times \mathbb{R}\}.$$

A bounded function a on the plane acts on $\mathfrak{h}$ by multiplication in y. In particular, 1_K is the orthogonal projection onto $\mathfrak{h}_K$.

We use the Gaussian bounds (4) and (5); Hölder's inequality also puts every finite product of fields in $L^2(\mu)$. Write $U(v) = \hat{\tau}_v$ for planar translations. Symmetries act unitarily and multiplicatively on $L^2(\mu)$ and commute with conjugation; Θ_1 is an antiunitary involution. The reconstructed scalar product was defined in section 1, and

$$\langle \Omega_1, [\Phi]_1 \rangle = \mu(\Phi), \qquad \|[\Phi]_1\| \le \|\Phi\|_{L^2(\mu)} \quad (\Phi \in L^2(\Sigma_1^+, \mu)). \tag{28}$$

All scalar products are antilinear in their first argument.

4.1. Localization to one pair.

For $d > 0$ and $z \in \mathbb{R}^2$ let

$$\varphi_{d,z}(y) = (\pi d^2)^{-1/2} 1_{B(z,d)}(y).$$

The normalization and covariance identities are

$$\|\varphi_{d,z}\|_2 = 1, \qquad \int \varphi_{d,z}(y)^2 \, \mathrm{d}z = 1, \qquad \int \|\varphi_{d,z} f\|^2 \, \mathrm{d}z = \|f\|^2, \tag{29}$$

$$\varphi_{d,z} \circ \theta_1 = \varphi_{d,\theta_1 z}, \qquad \varphi_{d,z} \circ R_{-t} = \varphi_{d,R_t z}, \qquad \tau_v \varphi_{d,z} = \varphi_{d,z+v}, \qquad R_t \cdot (\varphi_{d,z} f) = \varphi_{d,R_t z}(R_t \cdot f). \tag{30}$$

The third identity in (29) follows from the second by Tonelli's theorem. Both $z \mapsto \varphi_{d,z}$ in $L^2(\mathbb{R}^2)$ and $z \mapsto \varphi_{d,z} f$ in $\mathfrak{h}$ are norm continuous by dominated convergence, since the boundary circle has measure zero.

Lemma 4.1 (Localization of a bilinear form)**.** Let $K_1, K_2 \subset \mathbb{R}^2$ be Borel, and let $\beta : \mathfrak{h}_{K_1} \times \mathfrak{h}_{K_2} \to \mathbb{C}$ be bounded and bilinear. Suppose $\beta(f_1, f_2) = 0$ whenever $f_a \in \mathfrak{h}_{K_a \cap A_a}$ for two disjoint compact sets $A_1, A_2 \subset \mathbb{R}^2$. Then, for every Borel set A,

$$\beta(1_A f_1, 1_A f_2) = \beta(1_A f_1, f_2), \tag{31}$$

and, for every $d > 0$,

$$\beta(f_1, f_2) = \int \beta(\varphi_{d,z} f_1, \varphi_{d,z} f_2)\, \mathrm{d}z. \tag{32}$$

The integral in (32) is absolutely convergent. ■

Proof. The assumed vanishing holds also for two disjoint Borel sets A, A'. To see this, exhaust each by increasing compact subsets up to a null set, using inner regularity on its intersections with bounded discs. The corresponding multiplications converge strongly on $\mathfrak{h}$, so continuity of β gives $\beta(1_A f_1, 1_{A'} f_2) = 0$. Splitting $f_2 = 1_A f_2 + 1_{A^c} f_2$ proves (31).

For fixed f_2, the Riesz representation theorem gives $u \in \mathfrak{h}_{K_1}$ such that $\beta(f, f_2) = \int \overline{u(x)} \cdot f(x)\, \mathrm{d}x$. By (31),

$$\beta(\varphi_{d,z} f_1, \varphi_{d,z} f_2) = \int \varphi_{d,z}(y)^2 \overline{u(x)} \cdot f_1(x)\, \mathrm{d}x.$$

Since $\overline{u} \cdot f_1$ is integrable, Fubini's theorem and (29) prove both assertions about the integral. □

We use Hilbert-space integrals in their weak sense. If $V(z)$ is norm continuous with values in a Hilbert space and $\int |a(z)| \|V(z)\|\, \mathrm{d}z < \infty$, its integral is the unique vector v satisfying

$$\langle \eta, v \rangle = \int a(z) \langle \eta, V(z) \rangle\, \mathrm{d}z.$$

Existence follows from the Riesz theorem [2, Theorem 3.3], and $\|v\| \le \int |a| \|V\|$. Bounded linear maps pass through this integral; an antiunitary involution J gives $Jv = \int \overline{a}\, JV$. An integral of vectors in a closed subspace belongs to that subspace, by orthogonality to its complement. In $L^2(\mu)$ we shall repeatedly use

$$\mu\left(\Phi \int aV\right) = \int a(z) \mu(\Phi V(z))\, \mathrm{d}z \qquad (\Phi \in L^2(\mu)). \tag{33}$$

For complex angles we use the complex linear rotation R_ζ defined by $(R_\zeta \xi)^\pm = e^{\pm i\zeta} \xi^\pm$, where $\xi^\pm = \xi^0 \pm i\xi^1$. Thus $R_{i\beta}(\ell, 0) = (\ell \cosh\beta, i\ell \sinh\beta)$.

4.2. **Bounded local fields have only vacuum classes.** The limiting pair will be described by a bounded map $X : L^1(\mathbb{R}^2; \mathbb{C}) \to L^2(\mu)$. Write L^1_+ for the functions vanishing a.e. on $\{y^0 < 0\}$ and L^1_- for those vanishing a.e. on $\{y^0 > 0\}$. We require, for some $C \ge 0$, $s \in \{1, -1\}$ and $w \in \mathbb{Z}$, that

$$X \text{ is complex linear}, \qquad \|X(h)\|_{L^2(\mu)} \le C \|h\|_1, \tag{34}$$

$$U(v) X(h) = X(\tau_v h) \quad (v \in \mathbb{R}^2,\ h \in L^1), \tag{35}$$

$$X(h) \in L^2(\Sigma_1^+, \mu) \quad (h \in L^1_+), \tag{36}$$

$$\Theta_1 X(h) = s X(\overline{h} \circ \theta_1), \qquad \hat{R} X(h) = (-i)^w X(h \circ R^{-1}) \quad (h \in L^1). \tag{37}$$

Here $(\tau_v h)(y) = h(y - v)$.

Lemma 4.2 (Vacuum classes)**.** Assume (S), and let X satisfy (34)–(37). There is a vector $v_0 \in \ker H_1$ such that

$$[X(h)]_1 = \left(\int h\right) v_0 \qquad (h \in L^1_+).$$

■

Proof. Let E_1 be the joint spectral measure of the reconstruction and put

$$\mathcal{W} = \{\xi \in \mathbb{C}^2 : \Re\xi^0 > |\Im\xi^1|\}, \qquad \mathsf{T}(\xi) = \int e^{-\xi^0\omega + i\xi^1 p_2}\, \mathrm{d}E_1(\omega, p_\perp).$$

The cone condition means that E_1 is carried by $\{\omega \geq |p_2|\}$. The cone semigroup is therefore a contraction on $\overline{\mathcal{W}}$, strongly continuous there and holomorphic in operator norm on $\mathcal{W}$. It satisfies

$$\mathsf{T}(\xi)\mathsf{T}(\xi') = \mathsf{T}(\xi + \xi'), \qquad \mathsf{T}(\xi)^* = \mathsf{T}(\xi^*), \quad \xi^* = (\overline{\xi^0}, -\overline{\xi^1}),$$

and $\mathsf{T}(y)[\Phi]_1 = [U(y)\Phi]_1$ for real y with $y^0 \geq 0$. These conclusions follow from the spectral-calculus argument in [5, Section 8.2]: the multiplier has modulus at most one on the closed tube, and on a smaller tube its first two derivatives are bounded by constants times $\omega e^{-\varepsilon\omega}$ and $\omega^2 e^{-\varepsilon\omega}$. Dominated convergence gives strong continuity, and Taylor's formula gives operator-norm holomorphy.

Set $\psi(h) = [X(h)]_1$ for $h \in L^1_+$. The contraction in (28) and the bound (34) give the first assertion below. The semigroup identity just stated, together with (35), gives the second:

$$\|\psi(h)\| \leq C\|h\|_1, \qquad \mathsf{T}(y)\psi(h) = \psi(\tau_y h) \quad (y^0 \geq 0). \tag{38}$$

Take $g_n = n^2 1_{(0,1/n)^2}$. The vectors $\psi(g_n)$ are bounded, so a subsequence converges weakly to a vector v_0 with $\|v_0\| \leq C$. Apply diagonal extraction in the separable closed span of this bounded sequence and then the Riesz theorem [2, Proposition 3.13 and Lemma 3.11]; no separability of the whole Hilbert space is needed.

For $h \in L^1_+$ we claim

$$\langle \eta, \psi(h)\rangle = \int h(y)\langle \eta, \mathsf{T}(y)v_0\rangle\, \mathrm{d}y. \tag{39}$$

Indeed,

$$\|h * g_n - h\|_1 \leq \sup_{|a|\leq 2/n} \|\tau_a h - h\|_1 \longrightarrow 0.$$

For fixed η, the bounded functional $k \mapsto \langle \eta, \psi(k)\rangle$ on L^1_+ is integration against a bounded function. Fubini's theorem, followed by (38), consequently gives

$$\langle \eta, \psi(h * g_n)\rangle = \int h(y)\langle \mathsf{T}(y)^*\eta, \psi(g_n)\rangle\, \mathrm{d}y.$$

The integrand is bounded by $C\|\eta\||h(y)|$. Weak convergence on the right, norm convergence on the left and dominated convergence prove (39).

To locate the spectral support of v_0, consider the bounded function

$$L(\xi) = \langle v_0, \mathsf{T}(\xi)v_0\rangle \quad (\xi \in \overline{\mathcal{W}}), \qquad |L| \leq C^2.$$

It is continuous on $\overline{\mathcal{W}}$ and holomorphic on $\mathcal{W}$. The semigroup identity and (39) yield

$$\langle \psi(h), \psi(k)\rangle = \iint \overline{h(y)}k(y')L(y^* + y')\, \mathrm{d}y\, \mathrm{d}y' \qquad (h, k \in L^1_+). \tag{40}$$

All integrals converge absolutely by the bound on L.

Write $\mathfrak{B}(h,k) = \mu(X(h)X(k))$. If h vanishes above a line $y^0 = c$ and k vanishes below that line, then

$$\mathfrak{B}(h,k) = s\iint h(u)k(y)L(y-u)\, \mathrm{d}u\, \mathrm{d}y. \tag{41}$$

Translations reduce this to $c = 0$. For that case let $\widetilde{h} = \overline{h} \circ \theta_1 \in L^1_+$. The reflection identity gives $X(h) = s\Theta_1 X(\widetilde{h})$; apply (2) and (40), then change variables $y' = \theta_1 u$. Since $(\theta_1 u)^* = -u$, this gives (41).

We next use the quarter turn to continue L between the four time directions. Let $\epsilon = (-1)^w$. The covariance of X gives

$$\mathfrak{B}(h,k) = \epsilon\, \mathfrak{B}(h \circ R^{-1}, k \circ R^{-1}). \tag{42}$$

If $y_0^0 > 0$ and $y_0^1 < 0$, choose $0 < r < \frac{1}{2}\min(y_0^0, -y_0^1)$ and set

$$h = (\pi r^2)^{-1} 1_{B(0,r)}, \qquad k = (\pi r^2)^{-1} 1_{B(y_0,r)}.$$

These functions have integral one. Both this pair and its quarter turn are ordered across a horizontal line. Apply (41) to the two sides of (42). Their product measure has mass one and concentrates on differences $y - u = y_0$ as $r \downarrow 0$. Continuity of L gives $L(Ry_0) = \epsilon L(y_0)$. The domain $\mathcal{W} \cap R^{-1}\mathcal{W}$ is open, convex and contains this real quadrant. The holomorphic identity theorem therefore gives

$$L(R\xi) = \epsilon L(\xi) \qquad (\xi \in \mathcal{W} \cap R^{-1}\mathcal{W}). \tag{43}$$

To use the bound on L, fix $\ell > 0$ and put $y_\ell = (\ell, 0)$. On $R^{-m}\mathcal{W}$ define $L_m(\xi) = \epsilon^m L(R^m\xi)$, $m \in \mathbb{Z}$. Equation (43) makes consecutive functions agree on their overlaps, and $L_{m+4} = L_m$. If $\zeta = t + i\beta$, then

$$R_\zeta y_\ell \in R^{-m}\mathcal{W} \quad \Longleftrightarrow \quad \cos(t + m\pi/2) > 0,$$

because the real parts of its two light-cone coordinates after applying R^m are $\ell e^{\mp\beta}\cos(t + m\pi/2)$. These sets of angles cover $\mathbb{C}$. Opposite ones do not meet, and on each other overlap the equality of consecutive L_m applies. Thus $L_m(R_\zeta y_\ell)$ defines one entire function of ζ, bounded by C^2. Liouville's theorem [1, Theorem 6.16] makes it constant. In particular,

$$L(R_{i\beta} y_\ell) = L(y_\ell) \qquad (\beta \in \mathbb{R},\ \ell > 0). \tag{44}$$

Let $\varrho = \langle v_0, E_1(\cdot)v_0\rangle$. It is a finite positive measure on $\{\omega \geq |p_2|\}$, and (44) reads

$$\int e^{-\ell(\omega\cosh\beta + p_2\sinh\beta)}\,\mathrm{d}\varrho = \int e^{-\ell\omega}\,\mathrm{d}\varrho. \tag{45}$$

On this cone

$$\omega\cosh\beta + p_2\sinh\beta = \tfrac{1}{2}(\omega + p_2)e^{\beta} + \tfrac{1}{2}(\omega - p_2)e^{-\beta} \geq 0.$$

The integrand on the left of (45) is at most one. As $\beta \to +\infty$ it tends to the indicator of $\{\omega + p_2 = 0\}$. With $\ell > 0$ fixed, dominated convergence gives

$$\varrho\{\omega + p_2 = 0\} = \int e^{-\ell\omega}\,d\varrho.$$

Now let $\ell \downarrow 0$. Monotone convergence makes the right side tend to the total mass of ϱ, so this set carries all of the measure. Letting $\beta \to -\infty$ gives the same assertion for $\{\omega - p_2 = 0\}$. Hence ϱ is carried by $\{\omega = p_2 = 0\}$.

The spectral theorem now gives $v_0 \in \operatorname{Dom} H_1$ and $\|H_1 v_0\|^2 = \int \omega^2\,\mathrm{d}\varrho = 0$. It also gives $\mathsf{T}(y)v_0 = v_0$ for real $y^0 \geq 0$, since

$$\|\mathsf{T}(y)v_0 - v_0\|^2 = \int |e^{-y^0\omega + iy^1 p_2} - 1|^2\,\mathrm{d}\varrho = 0.$$

Substitute this into (39) to conclude. □

4.3. **From a vacuum class to a constant random variable.** The preceding lemma is an assertion in the reconstructed Hilbert space. The contact argument requires an assertion in $L^2(\mu)$. The L^1 bound on X supplies this stronger conclusion.

Lemma 4.3 (Constancy of bounded local fields)**.** Assume (S) and $\ker H_1 = \mathbb{C}\Omega_1$. Every map X satisfying (34)–(37) obeys

$$X(h) = \mu(X(h))\,1 \quad \text{in } L^2(\mu) \qquad (h \in L^1(\mathbb{R}^2)).$$

■

Proof. The conjugate map $X^{\mathrm{c}}(h) = \overline{X(\overline{h})}$ satisfies the same conditions with weight $-w$. For translations and the quarter turn this follows by commuting their operators with conjugation; the quarter-turn phase becomes $\overline{(-i)^w} = (-i)^{-w}$. For the reflection use that Θ_1 commutes with conjugation. The bound and positive-side measurability are preserved as well. Lemma 4.2 therefore gives

$$[X(h)]_1,\ [X^{\mathrm{c}}(h)]_1 \in \mathbb{C}\Omega_1 \qquad (h \in L^1_+).$$

If $[\Phi]_1 = a\Omega_1$ and $[\Psi]_1 = b\Omega_1$, then (28) identifies $a = \mu(\Phi)$ and $b = \mu(\Psi)$. The reflected scalar product (2) then gives

$$\mu((\Theta_1\Phi)\Psi) = \mu(\Theta_1\Phi)\mu(\Psi). \tag{46}$$

Consider the covariance

$$E(h,k) = \mu(\overline{X(h)}X(k)) - \overline{\mu(X(h))}\mu(X(k)).$$

It is sesquilinear, translation invariant, satisfies $E(k,h) = \overline{E(h,k)}$, and obeys

$$|E(h,k)| \le 2C^2\|h\|_1\|k\|_1. \tag{47}$$

For $h \in L^1_-$ and $k \in L^1_+$ put $\Phi = sX^{\mathrm{c}}(h \circ \theta_1)$ and $\Psi = X(k)$. Both have vacuum classes, and the reflection rule gives $\Theta_1\Phi = \overline{X(h)}$. Thus (46) says $E(h,k) = 0$. Translations and Hermitian symmetry show more generally that $E(h,k) = 0$ whenever h and k lie on opposite sides of any line $y^0 = c$.

For arbitrary $h, k \in L^1$ partition the plane into strips $A_j = \{j\delta \le y^0 < (j+1)\delta\}$. The decompositions into these strips converge in L^1, and (47) makes the resulting double series for $E(h,k)$ absolutely convergent. Every term with different strip indices vanishes by the preceding paragraph. Hence

$$|E(h,k)| \le 2C^2\|h\|_1 \sup_j \|k1_{A_j}\|_1.$$

Set $G(a) = \int |k(a,y^1)|\,\mathrm{d}y^1 \in L^1(\mathbb{R})$. For $M > 0$,

$$\|k1_{A_j}\|_1 \le M\delta + \int G1_{\{G>M\}}.$$

First take M large and then δ small. The supremum therefore tends to zero, so $E = 0$. Its diagonal value is $E(h,h) = \|X(h) - \mu(X(h))1\|^2_{L^2(\mu)}$, proving the lemma. □

4.4. **Limits of localized pairs.** Fix labels $\lambda_a = (\sigma_a, \chi_a)$, $a = 1, 2$, with $\chi_a \in L^2(\mathbb{R};\mathbb{R})$, and put

$$w = w(\sigma_1) + w(\sigma_2) \in \{-2,-1,0,1,2\}, \qquad s = s(\sigma_1)s(\sigma_2), \qquad C_\lambda = 2c_4^2\|\chi_1\|_2\|\chi_2\|_2.$$

Recall that $\mathbb{B}_\lambda(\alpha) = \mathbb{B}(\alpha \otimes \lambda)$, where $(\alpha \otimes \lambda)(x) = \alpha(y)\chi(x_3)e^\sigma$. Define

$$\Pi_d(z) = \mathbb{B}_{\lambda_1}(\varphi_{d,z})\mathbb{B}_{\lambda_2}(\varphi_{d,z}), \qquad X_d(g) = \int g(z)\Pi_d(z)\,\mathrm{d}z. \tag{48}$$

By (5) and $\|\varphi_{d,z} \otimes \lambda_a\| \le \sqrt{2}\|\chi_a\|_2$, the map Π_d is continuous with

$$\|\Pi_d(z)\|_{L^2(\mu)} \le C_\lambda, \qquad \|X_d(g)\|_{L^2(\mu)} \le C_\lambda\|g\|_1. \tag{49}$$

In particular, the integral is well defined for every $g \in L^1$.

Lemma 4.4 (Localized pair fields)**.** Assume (G), (T) and (Q_{12}). The translation and quarter-turn identities below and the mean formula hold under these assumptions. If (R_1) holds as well, so does the reflection identity:

$$U(v)X_d(g) = X_d(\tau_v g), \qquad \Theta_1 X_d(g) = sX_d(\overline{g} \circ \theta_1), \qquad \hat{R}X_d(g) = (-i)^w X_d(g \circ R^{-1}), \tag{50}$$

$$\mu(X_d(g)) = \kappa_d \int g, \qquad \kappa_d = \mu(\Pi_d(0)), \qquad |\kappa_d| \le C_\lambda. \tag{51}$$

If $w \neq 0$, then $\kappa_d = 0$. For the remaining assertions assume also (R_1) and (RP_1). If g vanishes on $\{z^0 < d\}$, then $X_d(g) \in L^2(\Sigma_1^+, \mu)$.

For the fixed labels λ_1, λ_2, every sequence $d_n \downarrow 0$ has a subsequence, common to all $g \in L^1$, along which $X_{d_n}(g)$ converges weakly in $L^2(\mu)$. Its limit satisfies (34)–(37) with constant C_λ, sign s and weight w.

If in addition (S) holds and $\ker H_1 = \mathbb{C}\Omega_1$, then, for all $g \in L^1$ and $\Phi \in L^2(\mu)$,

$$\mu(\Phi X_d(g)) - \mu(\Phi)\mu(X_d(g)) \longrightarrow 0 \qquad (d \downarrow 0). \tag{52}$$

■

Proof. The field actions and (30) give

$$U(v)\Pi_d(z) = \Pi_d(z+v), \qquad \Theta_1 \Pi_d(z) = s\Pi_d(\theta_1 z), \qquad \hat{R}\Pi_d(z) = (-i)^w \Pi_d(Rz).$$

The reflection formula uses that each $\varphi_{d,z}$ is real; the quarter-turn formula uses $R \cdot (\alpha \otimes \lambda) = (-i)^{w(\sigma)}(\alpha \circ R^{-1}) \otimes \lambda$. Pass these operators through the integral in (48), conjugating the scalar coefficient for Θ_1, and change variables. This proves (50). Translation invariance makes $\mu(\Pi_d(z))$ independent of z, proving (51); the bound follows from (49). At $z = 0$ the quarter-turn identity gives $\kappa_d = (-i)^w \kappa_d$. Since w is a pair weight, $(-i)^w \neq 1$ for $w \neq 0$, so $\kappa_d = 0$ in that case. For $z^0 \geq d$ the disc $B(z, d)$ lies in the positive half-plane. Each field in $\Pi_d(z)$ is then measurable on the positive side and belongs to every finite $L^p(\mu)$; their product belongs to $L^2(\Sigma_1^+, \mu)$. The integral stays in this closed subspace.

For the weak limits, choose a countable dense set $\{g_j\}$ in L^1, for example finite sums of dyadic-square indicators with coefficients in $\mathbb{Q} + i\mathbb{Q}$. The bound (49) and successive weak subsequence extractions give a single subsequence along which every $X_{d_n}(g_j)$ converges weakly. For arbitrary $g \in L^1$ and $\Psi \in L^2(\mu)$,

$$\begin{aligned} |\langle \Psi, X_{d_n}(g) - X_{d_m}(g)\rangle| &\leq 2C_\lambda \|\Psi\| \|g - g_j\|_1 \\ &\quad + |\langle \Psi, X_{d_n}(g_j) - X_{d_m}(g_j)\rangle|. \end{aligned}$$

The scalar pairings are therefore Cauchy. Their limits define, by the Riesz theorem, a vector $X(g)$ with $\|X(g)\| \leq C_\lambda \|g\|_1$. Linearity and the covariance identities pass to the limit. This is also true for the antiunitary Θ_1, since $\langle \Psi, \Theta_1 v\rangle = \overline{\langle \Theta_1 \Psi, v\rangle}$.

It remains to check positive-side measurability of $X(g)$ when $g \in L^1_+$. If g vanishes on $\{z^0 < \delta\}$ for some $\delta > 0$, then $X_{d_n}(g) \in L^2(\Sigma_1^+, \mu)$ once $d_n < \delta$. A closed subspace of a Hilbert space is weakly closed, so the same holds for $X(g)$. For general $g \in L^1_+$, approximate it by $g1_{\{z^0 \geq \delta\}}$ in L^1 and use the bound on X. Thus all the conditions on the limiting field hold.

Finally, suppose (52) fails for fixed g, Φ. There are $\delta > 0$ and $d_n \downarrow 0$ for which the modulus of its left side is at least δ. Pass to the subsequence just constructed. Lemma 4.3 makes its limiting field constant: $X(g) = \mu(X(g))1$. Weak convergence, paired with $\overline{\Phi}$ and with 1, then sends that same left side to zero, a contradiction. □

4.5. **Pairs at contact.** For arbitrary $f_1, f_2 \in \mathfrak{h}$ define

$$Y_d(f_1, f_2) = \int \mathbb{B}(\varphi_{d,z} f_1)\mathbb{B}(\varphi_{d,z} f_2)\, \mathrm{d}z. \tag{53}$$

The next lemma compares this contact pair with the fields just studied.

Lemma 4.5 (Contact-pair estimates)**.** Assume (G). The integral (53) exists in $L^2(\mu)$, is bilinear in its entries, and obeys

$$\|Y_d(f_1, f_2)\|_{L^2(\mu)} \leq c_4^2 \|f_1\| \|f_2\|. \tag{54}$$

If $f_a = \alpha_a \otimes \lambda_a$ with $\alpha_a \in C_{\mathrm{c}}(\mathbb{R}^2)$, then

$$\|Y_d(f_1, f_2) - X_d(\alpha_1 \alpha_2)\|_{L^2(\mu)} \longrightarrow 0. \tag{55}$$

Under (T) and (Q_{12}), for arbitrary $f_1, f_2 \in \mathfrak{h}$ and $t \in \mathbb{R}$,

$$\mu(Y_d(R_t \cdot f_1, R_t \cdot f_2)) - \mu(Y_d(f_1, f_2)) \longrightarrow 0. \tag{56}$$

Under (S) and $\ker H_1 = \mathbb{C}\Omega_1$, for arbitrary $f_1, f_2 \in \mathfrak{h}$ and $\Phi \in L^2(\mu)$,

$$\mu(\Phi Y_d(f_1, f_2)) - \mu(\Phi)\mu(Y_d(f_1, f_2)) \longrightarrow 0. \tag{57}$$

All limits are as $d \downarrow 0$. ■

Proof. The integrand in (53) is norm continuous. By (5), Cauchy–Schwarz in z, and (29),

$$\int \|\mathbb{B}(\varphi_{d,z} f_1)\mathbb{B}(\varphi_{d,z} f_2)\|_{L^2(\mu)} \, \mathrm{d}z \le c_4^2 \int \|\varphi_{d,z} f_1\| \|\varphi_{d,z} f_2\| \, \mathrm{d}z \le c_4^2 \|f_1\| \|f_2\|.$$

This proves existence, bilinearity and (54). It also permits pairing the integral with any $\Phi \in L^2(\mu)$ as in (33).

For (55) let $\omega_a(d) = \sup_{|y-z| \le d} |\alpha_a(y) - \alpha_a(z)|$. Uniform continuity gives $\omega_a(d) \to 0$. Moreover,

$$\|\varphi_{d,z}\alpha_a - \alpha_a(z)\varphi_{d,z}\|_2 \le \omega_a(d), \qquad \|\varphi_{d,z}\alpha_a\|_2, \ \|\alpha_a(z)\varphi_{d,z}\|_2 \le \|\alpha_a\|_\infty.$$

Expand the difference of the two products of fields, changing one factor at a time, and apply (5). Its norm is at most

$$C_\lambda(\omega_1(d)\|\alpha_2\|_\infty + \|\alpha_1\|_\infty \omega_2(d)).$$

For $d \le 1$ it vanishes outside the fixed compact one-neighborhood K of $\operatorname{supp}\alpha_1 \cup \operatorname{supp}\alpha_2$. Integration therefore bounds the left side of (55) by $|K|$ times this expression, proving the limit.

The bilinear forms on the left of (56) are bounded uniformly in d by (54); those on the left of (57) have the same property for fixed Φ. It suffices to prove each limit on finite sums of plates with continuous compactly supported plane profiles. These form a dense subspace of $\mathfrak{h}$: first use totality of plates, then approximate each plane profile in L^2 by a continuous function of compact support. The extension follows, for any uniformly bounded family T_d, from

$$|T_d(f_1, f_2)| \le |T_d(f_1', f_2')| + M(\|f_1 - f_1'\|\|f_2\| + \|f_1'\|\|f_2 - f_2'\|).$$

We may therefore take $f_a = \alpha_a \otimes \lambda_a$ with continuous compactly supported α_a. The plate rotation formula, (55), and (51) give

$$\begin{aligned} &\mu(Y_d(R_t \cdot f_1, R_t \cdot f_2)) - \mu(Y_d(f_1, f_2)) \\ &\qquad = (e^{-iwt} - 1)\kappa_d \int \alpha_1 \alpha_2 + o(1). \end{aligned}$$

If $w = 0$, the phase difference vanishes. If $w \neq 0$, then $\kappa_d = 0$ by lemma 4.4. No convergence of κ_d as $d \downarrow 0$ is needed. This proves (56). Finally, (55) changes the expression in (57) by $o(1)$ when Y_d is replaced by $X_d(\alpha_1\alpha_2)$. The latter expression tends to zero by (52). Density completes both proofs. □

4.6. **Removing the disjointness conditions.** We now state the precise extension used in the proof of the main theorem. For $N \ge 0$ and $t \in \mathbb{R}$ let

$$D_N^t(f_1, \dots, f_N) = S_N(R_t \cdot f_1, \dots, R_t \cdot f_N) - S_N(f_1, \dots, f_N), \qquad S_0 = 1.$$

The Gaussian bounds imply that each D_N^t is bounded, symmetric and N-linear, and $D_0^t = D_1^t = 0$.

Proposition 4.6 (From disjoint supports to all entries)**.** Assume (S) and $\ker H_1 = \mathbb{C}\Omega_1$. Suppose that, for every $N \ge 2$ and $t \in \mathbb{R}$,

$$D_N^t(f_1, \dots, f_N) = 0$$

whenever $f_a \in \mathfrak{h}_{K_a}$ for compact pairwise disjoint $K_1, \dots, K_N \subset \mathbb{R}^2$. Then $D_N^t = 0$ for every $N \ge 0$ and every $t \in \mathbb{R}$. Consequently $(R_t)_*\mu = \mu$ for all t. ■

Proof. Fix t, write $D_N = D_N^t$, and suppose inductively that $D_{N-2} = 0$. The cases $N = 0, 1$ have already been noted. For a collection E of two-element subsets of $\{1, \dots, N\}$, let $\mathcal{P}(E)$ mean that D_N vanishes on entries $f_a \in \mathfrak{h}_{K_a}$ whenever the K_a are compact and $K_a \cap K_b = \varnothing$ for every $\{a, b\} \in E$. The assumed separated-support invariance gives $\mathcal{P}(E)$ when E contains every pair. We show that any one pair can be removed.

Suppose $\mathcal{P}(E)$ holds and $e = \{a, b\} \in E$. Fix compact sets K_c satisfying all the disjointness conditions in $E \setminus \{e\}$ and entries $f_c \in \mathfrak{h}_{K_c}$. With the other entries fixed, regard D_N as a bounded bilinear form β on $\mathfrak{h}_{K_a} \times \mathfrak{h}_{K_b}$. If its two arguments are further restricted over disjoint compact sets A, A', then the sets $K_a \cap A$, $K_b \cap A'$ and the remaining K_c satisfy every condition in E. Hence $\mathcal{P}(E)$ makes this value of β zero. Lemma 4.1 applies and gives, for all $d > 0$,

$$D_N(f_1, \dots, f_N) = \int \beta(\varphi_{d,z} f_a, \varphi_{d,z} f_b)\, \mathrm{d}z. \tag{58}$$

Put

$$\Phi = \prod_{c \neq a,b} \mathbb{B}(f_c), \qquad \Phi' = \prod_{c \neq a,b} \mathbb{B}(R_t \cdot f_c),$$

with empty products equal to one. Both belong to $L^2(\mu)$ by Gaussian domination. Using (30) in the rotated term of (58), and then using $R_t z$ as the integration variable in that term, yields

$$D_N(f_1, \dots, f_N) = \mu(Y_d(R_t \cdot f_a, R_t \cdot f_b)\Phi') - \mu(Y_d(f_a, f_b)\Phi). \tag{59}$$

Each term is integrable before this change of variables, by the integral estimate in lemma 4.5 and Cauchy–Schwarz in $L^2(\mu)$.

The induction hypothesis gives $\mu(\Phi') = \mu(\Phi)$. Subtract and add the corresponding products of means in (59):

$$\begin{aligned} D_N(f_1, \dots, f_N) = {} & \big[\mu(Y_d(R_t \cdot f_a, R_t \cdot f_b)\Phi') - \mu(Y_d(R_t \cdot f_a, R_t \cdot f_b))\mu(\Phi')\big] \\ & - \big[\mu(Y_d(f_a, f_b)\Phi) - \mu(Y_d(f_a, f_b))\mu(\Phi)\big] \\ & + \mu(\Phi)\big[\mu(Y_d(R_t \cdot f_a, R_t \cdot f_b)) - \mu(Y_d(f_a, f_b))\big]. \end{aligned}$$

As $d \downarrow 0$, the first two brackets vanish by (57); the third vanishes by (56). The left side is independent of d, so it is zero. This proves $\mathcal{P}(E \setminus \{e\})$.

Remove the pairs successively to obtain $\mathcal{P}(\varnothing)$. Thus D_N vanishes whenever its entries each lie over a compact subset of the plane, without any condition on intersections. These entries are dense in $\mathfrak{h}$, since $1_{\{|y| \le n\}} f \to f$ in L^2. Continuity gives $D_N = 0$ on $\mathfrak{h}^N$. Induction from $D_0 = 0$ for even N and from $D_1 = 0$ for odd N proves the assertion for all N and t.

In particular all moments of real Schwartz fields are invariant under R_t. The moment-determination consequence of Gaussian domination then gives $(R_t)_* \mu = \mu$. □

5. Ergodic components and the full Euclidean group

We complete the proof of theorem A. By lemma 1.1, the law has (H) and belongs to $\mathcal{C}_9$. The ergodic-component arguments in [5, Sections 7.3–7.5] apply under these hypotheses. For the translation-ergodic decomposition

$$\mu = \int \mu_\xi \, d\nu(\xi),$$

they give, outside a single ν-null set,

$$\mu_\xi \in \mathcal{C}_9, \qquad \mu_\xi \text{ satisfies } (G), \qquad \mu_\xi \text{ is invariant under } \mathbb{R}^3 \rtimes O_{\mathrm{cub}}, \qquad \ker H_1(\mu_\xi) = \mathbb{C}\Omega_1. \tag{60}$$

In particular each such component has (H). Neither this component theorem nor its hypotheses require closedness.

Fix one component with these properties. The cone theorem gives (S), and proposition 3.1 gives invariance under R_t on tests over pairwise disjoint compact planar sets, at every order. Proposition 4.6 therefore gives $(R_t)_*\mu_\xi = \mu_\xi$ for every $t \in \mathbb{R}$.

The orthogonal symmetries of μ_ξ form a group containing O_{cub} and all R_t^{12}. Conjugation by a cyclic permutation of the axes gives all R_t^{23}. Every matrix in $SO(3)$ is a product $R_{t_1}^{12}R_{t_2}^{23}R_{t_3}^{12}$: choose the first two rotations to send e_3 to its image under the matrix; the remaining factor fixes e_3 and is a rotation in its orthogonal plane. Since the cubic group also contains a reflection, this proves invariance under all of $O(3)$. It holds for every g on the same full-measure set of components; no intersection over an uncountable family of null sets is being taken.

For any measurable set A of distributions, the action is measurable because its evaluation on a test f is evaluation on $g^{-1} \cdot f$. Consequently

$$(g_*\mu)(A) = \int \mu_\xi(g^{-1}A)\, d\nu(\xi) = \int \mu_\xi(A)\, d\nu(\xi) = \mu(A).$$

Translation invariance gives the corresponding statement for every Euclidean isometry. Finally (4) and the action on $\mathbb{B}$ give moment covariance on all L^2 tests. This proves theorem A.

6. Diagonal positivity, the cone and closedness

The diagonal hypothesis cannot be replaced by the cone alone. We give explicit negative reflection forms for two laws that satisfy (H) and all ordered cone conditions (C_{ji}) but are not closed. We then give an invariant law positive at every plane for which closedness also fails. Together these examples distinguish the hypothesis of theorem A from both the cone and the Bianchi identity.

For real Schwartz vector tests and a bounded measurable, even, real symmetric positive-semidefinite matrix multiplier M, write

$$C_M(b, b') = (2\pi)^{-3} \int \widehat{b}(p)^\dagger M(p) \widehat{b'}(p)\, dp, \qquad \widehat{b}(p) = \int e^{-ip\cdot x} b(x)\, dx.$$

Let γ_M be the centered Gaussian law with characteristic functional $\gamma_M(e^{i\mathbb{B}(b)}) = e^{-C_M(b,b)/2}$. The Gaussian covariance criterion in [3, Appendix D, "Gaussian laws and reflection positivity"] gives existence, and $M \le I$ gives (G) by Plancherel's identity. The phases of a common translation cancel in C_M, so all these laws are translation invariant. For $g \in O(3)$, changing variables in the Fourier integral shows that $M(gp) = gM(p)g^T$ almost everywhere implies invariance under g. Closedness is equivalent to $M(p)p = 0$ almost everywhere: indeed,

$$C_M(\nabla u, \nabla u) = (2\pi)^{-3} \int |\widehat{u}(p)|^2 p^T M(p) p\, dp.$$

If $M(p)p = 0$ almost everywhere, the displayed integral vanishes for every u. Conversely, closedness makes it vanish for a Gaussian u, whose Fourier transform never vanishes; positivity of M then gives $M(p)p = 0$ almost everywhere. We also use the Gaussian reflection criterion from the same reference: if C_M is reflection invariant, its law is reflection positive precisely when $C_M(r \cdot b, b) \ge 0$ for every real smooth test compactly supported in the positive half-space.

6.1. A diagonal Maxwell covariance.

For $m \ge 0$, define

$$M^{\oplus}_{ab}(p) = \delta_{ab} \frac{|p|^2 - p_a^2}{|p|^2 + m^2} \quad (p \neq 0), \qquad M^{\oplus}(0) = 0, \qquad \gamma^{\oplus}_m = \gamma_{M^{\oplus}}.$$

This multiplier lies between zero and I and obeys $M^{\oplus}(gp) = gM^{\oplus}(p)g^T$ for signed permutations g. Thus the law has (G), translation invariance and cubic symmetry. To verify coordinate positivity, write $p = (p_1, k)$ and $\omega(k) = (|k|^2 + m^2)^{1/2}$. At the single point $k = 0$ when $m = 0$, set $\omega(0) = 1$;

this does not change any integral. The partial fraction decomposition is

$$M^{\oplus}(p) = \operatorname{diag}(0,1,1) + \frac{\operatorname{diag}(|k|^2, -p_2^2 - m^2, -p_3^2 - m^2)}{p_1^2 + \omega(k)^2}.$$

The residue matrix in lemma A.1 is $\operatorname{diag}(|k|^2, p_2^2 + m^2, p_3^2 + m^2) \geq 0$. That lemma gives (RP_1), and the spectral measures of all one-field vectors are supported on $\{(\omega, k) : \omega = (|k|^2 + m^2)^{1/2}\}$. The round cone $\{\omega \geq |k|\}$ is closed, contains zero and is stable under addition. Lemma A.2 therefore puts the whole reconstructed spectrum in this cone. Cubic symmetry gives all coordinate reflection positivities and all ordered cones (C_{ji}). Thus the law has (H). It is not closed: $p^T M^{\oplus}(p) p > 0$ whenever p has at least two nonzero components. The following test identifies its failure at every diagonal mirror.

Proposition 6.1. For every $m \geq 0$, the law $\gamma_m^{\oplus}$ fails reflection positivity at every plane $x_i = \sigma x_l + c$, $i < l$, $\sigma = \pm 1$, from either side. For a fixed plane and side one can use the same test forms for all $m \geq 0$. ∎

Proof. First take $c = 0$. Fix signs $\sigma, \nu \in \{1, -1\}$ and put

$$n = \frac{\nu}{\sqrt{2}}(e_i - \sigma e_l), \qquad H_+ = \{n \cdot x > 0\}, \qquad r = r_{n,0}.$$

The reflection exchanges e_i and σe_l and fixes the third basis vector. Its action on tests is $r \cdot b = -r\, b \circ r$. If $C_a(u', u) = C_{M^{\oplus}}(u' e_a, u e_a)$, the multiplier identities $M_{ll}^{\oplus}(rp) = M_{ii}^{\oplus}(p)$ and $M_{ii}^{\oplus}(rp) = M_{ll}^{\oplus}(p)$ give, for $b = u e_i + v e_l$,

$$C_{M^{\oplus}}(r \cdot b, r \cdot b) = C_{M^{\oplus}}(b, b), \qquad C_{M^{\oplus}}(r \cdot b, b) = -2\sigma C_i(v \circ r, u). \tag{61}$$

Between disjoint supports the scalar covariance has kernel

$$K_i(x) = (\partial_i^2 - m^2)\frac{e^{-m|x|}}{4\pi|x|}.$$

Indeed, write its multiplier as $1 - (p_i^2 + m^2)/(|p|^2 + m^2)$; the identity term vanishes between disjoint supports, and Fourier inversion of the Yukawa potential followed by integration by parts gives this formula. It remains valid for $m = 0$ by dominated convergence, since $|p|^{-2}$ is locally integrable in three dimensions. With $\rho = |x|$ and $s = m\rho$, direct differentiation yields

$$K_i(x) = \frac{e^{-s}}{4\pi\rho^3}\left[(s^2 + 3s + 3)\frac{x_i^2}{\rho^2} - (s^2 + s + 1)\right]. \tag{62}$$

In particular, since $s^2 + 3s + 3 \leq 3(s^2 + s + 1)$,

$$K_i(x) \leq -\frac{e^{-m\rho}}{16\pi\rho^3} < 0 \quad \text{if } x_i^2 \leq \rho^2/4. \tag{63}$$

Choose $\lambda > 0$ and a nonzero $\varphi \in C_c^{\infty}(\mathbb{R}^3)$ with $\varphi \geq 0$ and support in $\{|x| \leq \lambda/3\}$. Set

$$y_1 = \nu\sigma\lambda e_l, \qquad u(x) = \varphi(x + y_1), \qquad v(x) = -\sigma\varphi(rx - y_1), \qquad b = u e_i + v e_l.$$

The two supports lie in balls centered at $-\nu\sigma\lambda e_l$ and $\nu\lambda e_i$. Both centers have n-coordinate $\lambda/\sqrt{2}$, so both closed balls lie in H_+. The form $f = *^{-1} b$ is therefore an admissible test in (1); it does not depend on m.

By (61), the common variance is $V = 2C_i(\varphi, \varphi)$ and the reflected covariance is

$$J = 2\iint \varphi(y)\varphi(y') K_i(y - y' + 2\lambda e_l)\, dy\, dy'.$$

To obtain the displayed sign of the displacement from $2y_1$, use evenness of K_i and interchange y, y' if necessary. On the integration region,

$$|(y - y' + 2\lambda e_l)_i| \leq 2\lambda/3, \qquad 4\lambda/3 \leq |y - y' + 2\lambda e_l| \leq 8\lambda/3.$$

Thus (63) applies throughout, and

$$J \le -c\lambda^{-3}e^{-8m\lambda/3}\left(\int \varphi\right)^2 < 0 \tag{64}$$

with an absolute constant $c > 0$.

For Gaussian data $f, -f$ and coefficients $1, -1$, expansion of the four terms in (1) gives

$$\mathcal{Q}_{\gamma_m^{\oplus}}(n, 0; (1, -1); (f, -f)) = 4e^{-V}\sinh J < 0. \tag{65}$$

One may instead use tests $0, f$ and coefficients $1, -e^{V/2}$; their form is $e^J - 1 < 0$. This proves failure at the chosen plane and side. Translation of the tests carries these same values to every parallel plane, because $r_{n,c} = \tau_{cn} r_{n,0} \tau_{-cn}$ and the law is translation invariant. The choice of ν includes both sides. □

For radial φ, the witness also has an exact expression. Define

$$c_\varphi(m) = \int \varphi(y)\frac{\sinh(m|y|)}{m|y|}\,dy,$$

with quotient one at $m = 0$. Then

$$J = -c_\varphi(m)^2\frac{(4m^2\lambda^2 + 2m\lambda + 1)e^{-2m\lambda}}{16\pi\lambda^3}, \tag{66}$$

$$V = \frac{4}{3}(2\pi)^{-3}\int |\widehat{\varphi}(p)|^2\frac{|p|^2}{|p|^2 + m^2}\,dp. \tag{67}$$

To verify the first formula, a smooth solution of $\Delta G = m^2 G$ near a ball has spherical mean $G(0)\sinh(m\rho)/(m\rho)$. Green's formula gives the radial equation $S'' + 2\rho^{-1}S' = m^2 S$, with $S(0) = G(0)$; solving for ρS gives this mean. Radial integration therefore gives $\int \varphi G = c_\varphi(m)G(0)$. Apply it twice to the two variables in J, using $\Delta K_i = m^2 K_i$ away from the origin. Then $J = 2c_\varphi(m)^2 K_i(2\lambda e_l)$, which is (66). Symmetry among the three Fourier coordinates gives (67). At $m = 0$ these reduce to

$$J = -\frac{(\int \varphi)^2}{16\pi\lambda^3}, \qquad V = \frac{4}{3}\|\varphi\|_2^2.$$

6.2. **A mixture of white noises.** Let $D_1, D_2, D_3 \in (0, 1]$ be pairwise distinct. For each permutation $\pi \in \mathfrak{S}_3$, let $\gamma_{\pi D}$ be the white-noise law with variance $D_{\pi^{-1}(a)}$ in component a of $\mathbb{B}$, and put

$$m_1 = \frac{1}{6}\sum_{\pi\in\mathfrak{S}_3}\gamma_{\pi D}.$$

Each summand satisfies (G), is translation invariant and is invariant under coordinate reflections. The cubic group permutes the six summands. Opposite open half-spaces are independent for each white noise, since their covariance vanishes. Consequently, for positive-side $\Phi, \Psi \in L^2(m_1)$,

$$m_1((\Theta_j\Phi)\Psi) = \frac{1}{6}\sum_{\pi\in\mathfrak{S}_3}\overline{\gamma_{\pi D}(\Phi)}\gamma_{\pi D}(\Psi).$$

The inclusion $L^2(m_1) \subset L^2(\gamma_{\pi D})$ follows from $m_1 \ge \gamma_{\pi D}/6$. The displayed formula proves coordinate reflection positivity. It also identifies a reconstructed class by its six expectations, each unchanged by translations. Hence every reconstructed translation is the identity, so $H_j = 0$ and all transverse momenta vanish. This proves (H) and all ordered cones. Moreover

$$m_1(\mathbb{B}(\nabla u)^2) = \frac{D_1 + D_2 + D_3}{3}\|\nabla u\|_2^2 > 0$$

for every nonzero real Schwartz function u, so m_1 is not closed. Although the individual white noises have no correlations between opposite open half-spaces, their mixture need not be reflection positive at a reflection that permutes the variances.

Fix the plane, side and reflection from proposition 6.1. Take a nonzero real $\psi \in C_c^\infty(H_+)$ and set $*f_1 = \psi e_i$, $*f_2 = \psi e_l$. Let $\mathfrak{t}$ exchange i, l and write

$$\beta_a = e^{-D_a\|\psi\|_2^2/2}, \qquad Z_\pi = \beta_{\pi^{-1}(i)} - \beta_{\pi^{-1}(l)}.$$

For tests on the positive side, the white-noise characteristic functional factors between the two half spaces. Reflection permutes the weights by $\mathfrak{t}$, so its form for coefficients $(1,-1)$ is

$$\begin{aligned}\mathcal{Q}_{m_1}(n,0;(1,-1);(f_1,f_2)) &= \frac{1}{6}\sum_\pi \overline{Z_{\mathfrak{t}\pi}} Z_\pi \\ &= -\frac{1}{6}\sum_\pi Z_\pi^2 = -\frac{1}{3}\sum_{a<c}(\beta_a-\beta_c)^2 < 0. \end{aligned} \tag{68}$$

Here $Z_{\mathfrak{t}\pi} = -Z_\pi$, and the preimages of i, l run through all six ordered pairs of distinct indices. Distinct variances make the last inequality strict. The calculation works for both values of ν; translation gives the same failure at every parallel plane. Thus neither m_1 nor $\gamma_m^\oplus$ belongs to $\mathcal{C}_9$.

6.3. **An invariant nonclosed law.** The preceding examples fail the extra positivity hypothesis of theorem A. The following example satisfies it, and shows that the theorem applies to nonclosed laws.

Proposition 6.2 (An invariant nonclosed law)**.** For $m > 0$, let $M(p) = I - pp^T/(|p|^2 + m^2)$. The Gaussian law γ_M has (H), is invariant under every Euclidean isometry, and is reflection positive at every affine plane. It is not closed. ■

Proof. On $p^\perp$ the multiplier is I, and on $\mathbb{R}p$ its eigenvalue is $m^2/(|p|^2+m^2)$. Thus $0 \le M \le I$; it is measurable, even and real symmetric. The multiplier criteria give the Gaussian law, translation invariance and (G). Since

$$M(gp) = gM(p)g^T \quad (g \in O(3)), \qquad M(p)p = \frac{m^2}{|p|^2+m^2}p,$$

the law is orthogonally invariant and not closed.

For coordinate reflection positivity, split $p = (p_1, p_\perp)$ and put $\omega = (|p_\perp|^2 + m^2)^{1/2}$. Lemma A.1 applies to

$$M(p) = I + \frac{N(p)}{p_1^2+\omega^2}, \qquad N(p) = -pp^T.$$

Its polynomial part is constant, its numerator is polynomial in p_1, and $\omega \ge m > 0$. It obeys the required reflection symmetry $\theta_1 M(-p_1, p_\perp)\theta_1 = M(p)$. The residue matrix in that criterion, with $\mathsf{p} = (-i\omega, p_\perp)$, is

$$A = -\theta_1 N(-i\omega, p_\perp) = \overline{\mathsf{p}}\,\mathsf{p}^T, \qquad v^\dagger A v = |\mathsf{p}^T v|^2 \ge 0.$$

The criterion therefore gives (RP_1). Orthogonal and translation invariance transport it to every affine plane, including both sides. In particular all three coordinate conditions hold, so the law has (H). □

Closedness and diagonal positivity consequently play different roles. The cone and (H) alone do not give diagonal positivity, as (65) and (68) show. Diagonal positivity does give rotation invariance under (H), by theorem A, without forcing closedness, by proposition 6.2. The elimination of the extra two-point frequency by the Bianchi identity, recalled in eq. (20) and proved in [5, Section 8.5], still uses $dF = 0$. The examples above show why a conclusion of diagonal positivity from (H) and the cone requires an additional hypothesis.

Appendix A. Gaussian reflection forms and their spectral support

We record the two covariance calculations used in section 6. Write $p = (p_1, k)$, with $k \in \mathbb{R}^2$, and regard θ_1 as the matrix $\mathrm{diag}(-1, 1, 1)$. In the coordinate reconstruction, write $V_1(y) = \int e^{ik\cdot y}\, dE_1(\omega, k)$ for the transverse translation, with $y \in e_1^{\perp}$.

Lemma A.1 (A time-kernel formula)**.** Let M be bounded, measurable, even, real symmetric and positive semidefinite, and suppose $\theta_1 M(-p_1, k)\theta_1 = M(p_1, k)$. Suppose that, for almost every k, there are matrix polynomials Q_k, N_k and a measurable $\omega(k) > 0$ such that

$$M(p_1, k) = Q_k(p_1) + \frac{N_k(p_1)}{p_1^2 + \omega(k)^2}.$$

The coefficients of these polynomials are assumed measurable in k. Put $A(k) = -\theta_1 N_k(-i\omega(k))$. For a real $b \in C_c^\infty(\{x_1 > 0\}; \mathbb{R}^3)$ define

$$L_b(k) = \int_0^\infty e^{-\omega(k)s}\widehat{b}_\perp(s, k)\, ds, \qquad \widehat{b}_\perp(s, k) = \int_{\mathbb{R}^2} e^{-ik\cdot x_\perp} b(s, x_\perp)\, dx_\perp.$$

Then, for $t \geq 0$ and $y \in e_1^{\perp}$,

$$C_M(\theta_1 \cdot b, \tau_{te_1+y} \cdot b) = (2\pi)^{-2} \int_{\mathbb{R}^2} e^{-ik\cdot y} \frac{e^{-\omega(k)t}}{2\omega(k)} L_b(k)^\dagger A(k) L_b(k)\, dk. \tag{69}$$

If $A(k)$ is Hermitian and positive semidefinite almost everywhere, γ_M is reflection positive at $x_1 = 0$. Under this positivity hypothesis, the spectral measure of $[\mathbb{B}(b)]_1$ is the image under $k \mapsto (\omega(k), -k)$ of

$$(2\pi)^{-2} \frac{L_b(k)^\dagger A(k) L_b(k)}{2\omega(k)}\, dk.$$

■

Proof. For $h \in C_c^\infty((0, \infty))$ and a polynomial P, Fourier inversion and the inverse transform of $(p_1^2 + \omega^2)^{-1}$ give

$$\frac{1}{2\pi} \int \widehat{h}(p_1) P(p_1)\, dp_1 = 0,$$
$$\frac{1}{2\pi} \int \frac{\widehat{h}(p_1) P(p_1)}{p_1^2 + \omega^2}\, dp_1 = \frac{P(-i\omega)}{2\omega} \int_0^\infty e^{-\omega u} h(u)\, du.$$

For the first identity, derivatives of h vanish at zero. For the second, the case $P = 1$ follows from the inverse transform $e^{-\omega|u|}/(2\omega)$; the case $P = p_1$ follows by applying it to $-ih'$ and integrating by parts. Polynomial division by $p_1^2 + \omega^2$ gives the general case.

The transforms of the reflected and translated tests are $-\theta_1 \widehat{b}(-p_1, k)$ and $e^{-ip_1 t - ik\cdot y}\widehat{b}(p_1, k)$. For each pair of components, the product of their time transforms is the transform of

$$h_{ac}(u; k) = \int \overline{\widehat{b}_{\perp,a}(s, k)}\, \widehat{b}_{\perp,c}(u - s - t, k)\, ds.$$

This is smooth and compactly supported in $(0, \infty)$, because b is supported a positive distance from the mirror and $t \geq 0$. The full Fourier covariance is absolutely integrable, since M is bounded and the tests are Schwartz. Fubini permits integration first in p_1. Applying the two preceding identities entrywise removes Q_k and replaces $N_k(p_1)$ by $N_k(-i\omega(k))$. Substitution $u = s + s' + t$ factors the remaining integral into $e^{-\omega(k)t}\overline{L_{b,a}(k)} L_{b,c}(k)$ and proves (69).

At $t = 0$, $y = 0$ the integrand is nonnegative when $A \geq 0$. The Gaussian reflection criterion recalled in section 6 therefore gives reflection positivity. The same specialization shows that the displayed spectral measure is finite. The reconstruction identity identifies the left side of (69) with the Fourier–Laplace transform of the spectral measure of $[\mathbb{B}(b)]_1$. The right side is the transform of the stated image measure, so uniqueness identifies them. For completeness, uniqueness for finite

complex measures follows by first fixing integer $t \geq 0$ and applying Fourier uniqueness in y. After the change $s = e^{-\omega}$, equality of all these marginals gives equality against every polynomial in s on $[0, 1]$. Polynomial approximation gives equality against continuous functions; rectangles then determine the original measures. □

Lemma A.2 (From one-field vectors to the Gaussian reconstruction)**.** Let γ be a centered Gaussian law invariant under translations and reflection positive and invariant at $x_1 = 0$. Let $K \subset [0, \infty) \times \mathbb{R}^2$ be closed, contain $(0, 0)$ and satisfy $K + K \subset K$. If the spectral measure of $[\mathbb{B}(b)]_1$ is carried by K for every real smooth test compactly supported in $\{x_1 > 0\}$, then the whole joint spectral measure E_1 is carried by K. ■

Proof. Put $C(b, b') = \gamma(\mathbb{B}(b)\mathbb{B}(b'))$. For finitely many positive-side real tests b_a and coefficients z_a, let

$$\Phi = \sum_a z_a e^{i\mathbb{B}(b_a)}, \qquad z'_a = z_a e^{-C(b_a,b_a)/2}, \qquad \psi_a = [\mathbb{B}(b_a)]_1.$$

Gaussian integration and the reconstruction identity give

$$\langle [\Phi]_1, e^{-tH_1} V_1(y) [\Phi]_1 \rangle = \sum_{a,c} \overline{z'_a} z'_c \exp\big(C(\theta_1 \cdot b_a, \tau_{te_1+y} \cdot b_c)\big),$$

$$C(\theta_1 \cdot b_a, \tau_{te_1+y} \cdot b_c) = \int e^{-t\omega + ip\cdot y} \, dm_{ac}(\omega, p), \qquad m_{ac} = \langle \psi_a, E_1(\cdot)\psi_c \rangle.$$

The measures m_{ac} are finite and carried by K. Indeed, Cauchy–Schwarz on finite Borel partitions gives $|m_{ac}|(B) \leq [m_{aa}(B) m_{cc}(B)]^{1/2}$. On the additive semigroup $[0, \infty) \times \mathbb{R}^2$, the measure

$$\exp_*(m) = \sum_{r \geq 0} \frac{m^{*r}}{r!}, \qquad m^{*0} = \delta_{(0,0)},$$

converges in total variation, since $\|m^{*r}\| \leq \|m\|^r$. Its Fourier–Laplace transform is the exponential of that of m. The conditions on K ensure that $\exp_*(m_{ac})$ is carried by K. Transform uniqueness, as proved above, therefore identifies the spectral measure of $[\Phi]_1$ with

$$\sum_{a,c} \overline{z'_a} z'_c \exp_*(m_{ac}).$$

It follows that $E_1(K^c)[\Phi]_1 = 0$. The positive-side characters span a dense subspace of the positive-side L^2 space, and their classes are dense in the reconstruction; see [5, Section 7.1]. Thus $E_1(K^c) = 0$. □

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